

tional likelihood function reads

$$f^*(t) = \lambda^*(t) \exp \left\{ \int_{t_j}^t \lambda^*(t') dt' \right\}, \quad (1)$$

where λ^* is (locally) integrable and is known as the intensity function of the point process. Following (?), (?), we define the functional dependence of the intensity function to be given by a RNN with hidden state $\mathbf{h}_j \in \mathbb{R}^d$, where an exponential function guarantees that the intensity is *non-negative*

$$\lambda^*(t) = \exp \{ \mathbf{v}^t \cdot \mathbf{h}_j + w^t (t - t_j) + b^t \}. \quad (2)$$

Here the vector $\mathbf{v}^t \in \mathbb{R}^d$ and the scalars w^t and b^t are trainable variables. The update equation for the hidden variables of the recurrent network can be written as a general non-linear function

$$\mathbf{h}_j = f_\theta(t_j, \mathbf{X}_j, \mathbf{h}_{j-1}), \quad (3)$$

where t_j and $\mathbf{X}_j$ label the creation time and the text's BoW representation of review $\mathbf{r}_j$, respectively, and θ denotes the network's parameters. We thus use the BoW representation of the review text as marks in the recurrent marked temporal point process (?). Inserting Eq. (2) into Eq. (1) and integrating over time immediately yields the likelihood f^* as a function of $\mathbf{h}_j$.

Dynamic Neural Text Model: To model the text component of reviews we assume the words in review $\mathbf{r}_{j+1}$ are generated independently, conditioned on $\mathbf{h}_j$, the temporal representation of the RPP above. Specifically, we follow (?) and write the conditional probability of generating the i th word $\mathbf{w}_i^{j+1}$ of the $(j+1)$ th review as

$$p_\theta(\mathbf{w}_i^{j+1} | \mathbf{h}_j) = \frac{\exp \{ -z(\mathbf{w}_i^j, \mathbf{h}_j) \}}{\sum_{k=1}^V \exp \{ -z(\mathbf{w}_k^j, \mathbf{h}_j) \}}, \quad (4)$$

$$z(\mathbf{w}_i^j, \mathbf{h}_j) = -\mathbf{h}_j^T \mathbf{R} \mathbf{w}_i^j - \mathbf{b} \mathbf{w}_i^j, \quad (5)$$

where $\mathbf{R} \in \mathbb{R}^{d \times V}$ and $\mathbf{b} \in \mathbb{R}^V$ are trainable parameters, and $\mathbf{w}_i^j$ is the one-hot representation of the word at position i . The complete log-likelihood of the RPRM model can then be written as

$$\mathcal{L} = \sum_{a=1}^N \sum_{j=1}^{N_a} (\log f^*(\delta_{j+1}^a | \mathbf{h}_j) + \log P(\mathbf{X}_{j+1}^a | \mathbf{h}_j)), \quad (6)$$

where $\delta_{j+1}^a = t_{j+1}^a - t_j^a$ denotes the inter-review time for item a and $P(\mathbf{X}_{j+1}^a | \mathbf{h}_j)$ is a multinomial distribution over word probabilities (Eq. (4)) and counts.

Baseline model: In order to test our model we define LSTM-BoW, which models the inter-review time δ_{j+1} as the mean of an exponential distribution with parameter $\lambda_\phi(\mathbf{h}'_j)$, and the probability over words $p(\mathbf{w}_i^{j+1} | \mathbf{h}'_j)$ as $\text{softmax}(g_\phi(\mathbf{h}'_j))$. The functions λ_ϕ and g_ϕ are given by neural networks with parameter ϕ , and $\mathbf{h}'_j = f_\phi(t_j, \mathbf{X}_j, \mathbf{h}'_{j-1})$ is the hidden state of an LSTM network (?). We also consider additional LSTM and RPP models which only take t_j as input, as to check whether the BoW representation $\mathbf{X}_i$ helps in the prediction of the inter-review times.

Model	RMSE	R ²	Pred. Perplexity
LSTM	96.8813	0.1788	-
RPP	96.3794	0.1873	-
LSTM-BoW	95.3414	0.2046	519.90
RPRM	92.3850	0.2533	511.32

Table 1: Model performance on RMSE, R^2 and predictive perplexity.

Experiments and Results

We test our models on the Yelp19 dataset³. Specifically we take all reviews for businesses that are labeled with the *shopping* parent category from *01 Jan 2016* to *30 Nov 2018*. The creation time of a review is defined as the difference in days between the original timestamp and *01 Jan 2016*. Next, we group reviews by business. All businesses with less than 5 reviews are removed. The text from each review is converted into a BoW vector of size 2000 (?). The result from the preprocessing is a dataset that has in total 262193 reviews, 27185 businesses, 174122 users, 1910299 sentences and 13209813 words. On average we have 9.6 reviews per business with standard deviation of 22.8. Each review has on average 7.2 sentences and 50.2 words, with 5.9 and 46.14 standard deviation respectively. In the experiments, we randomly split each dataset into two parts: training set (80%) and test set (20%). We use grid search for hyper-parameters finding.

We trained all models on maximum likelihood and use two evaluation metrics: Root-mean-squared error (RMSE) on the inter-review times and predictive perplexity on the review text. The latter is defined as $\mathcal{PP} = \exp \left\{ -\frac{1}{T} \sum_{j=1}^T \frac{1}{|\mathcal{H}_j|} \sum_{i \in \mathbf{x}_j} \frac{\log p(\mathbf{w}_i^j | \mathbf{h}_{j-1})}{M_j} \right\}$ (?), where M_j is the number of words in review $\mathbf{r}_j$ and $|\mathcal{H}_j|$ is the number of reviews at time t_j . Our results are presented in Table 1 and show that the best model in both metrics is the Recurrent Point Review Model. Note also that the models that leverage the information encoded in the text (through $\mathbf{X}_j$) show improvement of the RMSE (with respect to the inter-review time) over the models which do not see $\mathbf{X}_j$.

Conclusion and Future Work

In this work we incorporate a bag of word language model as the marks of a recurrent temporal point process. This creates a model which characterize temporal and causal representation for text, allowing for a richer representation for costumers reviews. We show that this improves predictive performance for the time of the reviews, as well as opening the door for *text prediction*. We will extend this methodology for rating prediction as well as more complex models of text.⁴ Ratione fugiat in numquam, itaque sequi ut asperiores nostrum, est dolores voluptates vel, iste dolorum ad fugit consequuntur commodi dolore repellendus suscipit nostrum?Non

³<https://www.yelp.com/dataset>

⁴One part of this research has been funded by the Federal Ministry of Education and Research of Germany as part of the competence center for machine learning ML2R (01—S18038A).

sequi id ut optio officia reprehenderit exercitationem dolores
tenetur nulla, at odio iste repellendus sunt nihil facere? Ipsa
eaque velit, molestiae animi